



Customer Segmentation Analysis for Strategic Growth

Identifying and Retaining High-Value Retail Customers

Retaining high-value retail customers requires identifying distinct behavioral segments.

THE STRATEGIC IMPERATIVE

"To maximize high-value customer retention, we must transition from raw transaction data to actionable behavioral insights."

541,909 → **391,297**
Raw Records Cleaned Transactions

DATA METHODOLOGY & OBJECTIVE

The dataset initially contained 541,909 records, which we cleaned to 391,297 valid product transactions by removing missing IDs, duplicates, and non-product codes.

Our Strategic Goal:

- Identify distinct customer archetypes to enable highly targeted, segment-specific retention models.
- Leverage clean, high-value behavioral insights to transition standard shoppers into loyal advocates.



RFM analysis translates transaction history into actionable behavioral dimensions.

We engineered three core domain-derived features for each customer to form our baseline behavioral profiles:

Recency

The number of days since the customer's last purchase, anchored against the maximum invoice date.

Frequency

The total count of transactions per customer.

Monetary

The total spend calculated by multiplying quantity and unit price for each transaction line.



Log transformation successfully normalizes highly skewed retail transaction distributions.



METHODOLOGY AND RATIONALE

Initial data exploration revealed severe skewness in raw RFM values, a common issue in retail datasets.

To prepare the features for K-Means clustering—which is highly sensitive to outliers—we applied a log transformation.

This successfully reduced skewness across all dimensions:

-0.378920

RECENCY SKEWNESS

-0.009306

FREQUENCY SKEWNESS

0.401825

MONETARY SKEWNESS

K-Means clustering and PCA reveal three distinct customer behavioral segments.

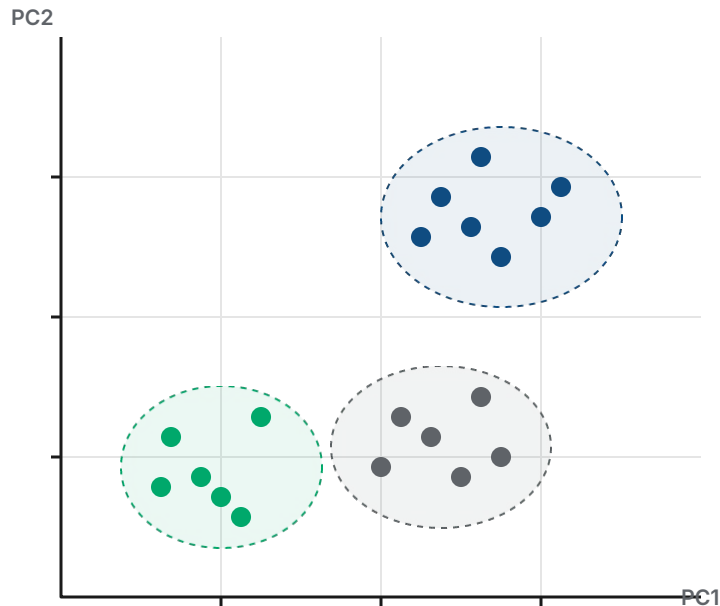
Methodology Overview

We utilized the K-Means algorithm to segment our standardized data. Using the Elbow Method, we determined that three clusters provided the optimal balance of granularity and interpretability.

To de-noise the data and visualize these segments, we applied Principal Component Analysis (PCA). The first two principal components successfully explain 92.23% of the total variance in the dataset, providing a clear 2D representation of our customer base.

92.23%

Total Variance Explained by PCA (PC1 & PC2)



2D Customer Projection via Principal Components

● Segment A ● Segment B ● Segment C



Logistic Regression confirms the clusters are highly linearly separable.

To validate the consistency of our K-Means segments, we trained supervised learning models to predict the cluster labels. We compared three architectures:

99.88%

LOGISTIC REGRESSION

Linear Model

98.38%

RANDOM FOREST

Ensemble Model

84.08%

DECISION TREE

Rule-Based Model

The near-perfect performance of Logistic Regression indicates that the segmented customer profiles are highly linearly separable within the log-scaled RFM space.



Three distinct customer segments emerged with vastly different engagement levels.

CHAMPIONS

1,014

CUSTOMERS

High frequency (256.04 avg) and massive monetary value (£6,254.51 avg) with recent engagement (16.10 days).

HIGH-VALUE CHAMPIONS

POTENTIAL LOYALISTS

1,842

CUSTOMERS

The middle tier with consistent purchasing (59.44 avg frequency, £1,077.76 avg spend).

RELIABLE REGULARS

AT-RISK / LAPSED

1,478

CUSTOMERS

Characterized by long inactivity (178.64 days avg recency) and lower overall spend (£289.55 avg).

RE-ENGAGEMENT TARGETS



Purchasing behaviors differ significantly across customer segments.

Product affinity analysis reveals distinct purchasing preferences for each segment, informing targeted marketing:

Champions

Favor high-volume decorative items.

Top Products

- White Hanging Heart T-Light Holder
- Medium Ceramic Top Storage Jar

Potential Loyalists

Share some product preferences with Champions.

Top Products

- Jumbo Bag Red Retrospot

At-Risk / Lapsed

Historically purchased smaller, lower-value items.

Top Products

- Essential Balm 3.5g Tin
- Brocade Ring Purse



Strategic Recommendations

Based on the segment profiles, we recommend three targeted strategies to maximize high-value retention:

Tailored strategies will drive retention, growth, and re-engagement across segments.



RETENTION Champions

Implement a **VIP loyalty program** offering early access to new product launches to sustain their high transaction frequency.



GROWTH Potential Loyalists

Deploy **cross-selling tactics**. Bundle offers for top items like Ceramic Jars to increase their lifetime value.



RE-ENGAGEMENT At-Risk Customers

Launch targeted **win-back email campaigns** featuring deep discounts on high-margin items to incentivize a return purchase.



Real-world constraints require careful contextualization of the segmentation model.



Class Imbalance

The massive 'At-Risk' majority versus the dense 'Champions' minority reflects real-world Pareto distributions, which can bias marketing resource allocation.



Temporal Overfitting (Data Leakage)

Anchoring the snapshot date to the maximum invoice date risks look-ahead bias if deployed on live streaming data.



Geographic Skew

The dataset is predominantly driven by UK wholesale/retail transactions. Direct application to international markets requires country-wise normalization.



Frequency-Weighted Scaling

Prevents high-spend features from dominating the clusters

Technical Motivation

To prevent the 'Monetary' feature from dominating the K-Means distance metrics, we implemented manual feature-weight scaling. This adjustment balances transaction frequency against currency-heavy features, ensuring a more holistic behavioral profile.

We applied the following specific weights to the standardized data before re-running the clustering algorithm:

Recency

1.0



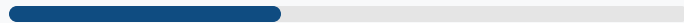
Frequency

1.5



Monetary

0.8



Data-Driven Segmentation Roadmap

Data-driven segmentation provides a clear roadmap for maximizing customer lifetime value.

METHODOLOGY & SEGMENTS

Our transition from raw data engineering to unsupervised K-Means clustering has successfully isolated our most valuable retail customers.

By understanding the distinct RFM profiles and product affinities of our Champions, Potential Loyalists, and At-Risk segments, we can transition from generic marketing to targeted, high-impact campaigns.

STRATEGIC INITIATIVES



VIP Loyalty & Retention: Implementing our recommended VIP loyalty programs and frequency-weighted cross-selling initiatives...



Cross-Selling Growth: ...will directly maximize high-value customer retention and drive sustainable strategic growth.

